

CONFETTI: CONvolutional FibEr TracT Inference with Multi-Resolution Graphs

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Abstract. Per-tract-bundle mean summaries are commonly used for diffusion MRI based white-matter (WM) microstructure analyses to reduce the dimensionality of the data, but collapsing each tract to a set of scalars (one per diffusion metric) discards the within-tract spatial structure and vastly reduces spatial localization of effects. We introduce **CONFETTI** (CONvolutional FibEr TracT Inference), a WM analysis pipeline built around three components: (i) a hierarchical multi-resolution neighborhood graph over parametrized fiber axes that retains within-tract spatial structure and between-tract proximity; (ii) an implicit-neural-representation-based imputation for missing diffusion tract profiles; and (iii) a Graph Convolutional Network (GCN) analysis including per-tract/node/covariate interpretable attribution. We evaluate CONFETTI with three structurally distinct architectures: 1) single-level GCN, 2) multi-scale parallel-branch concatenation, 3) hierarchical Graph U-Net, in a study of WM microstructure in infants with Down Syndrome (DS). We investigate DS classification as well as prediction of later cognitive and behavioral outcomes. We compare CONFETTI against per-tract mean baselines (logistic regression and XGBoost). Our results show that while the classification performance is comparable between baselines and the three GCN models, the attribution results show a spatially finer pattern, as well as the emergence of the anterior commissure (AC) as a tract of higher relevance in DS. It is the top-ranked tract for the GCN-based models, while neither baseline lists it in its highly ranked tracts. Via several follow-up experiments, we show that this spatial finding is robust, and that it requires CONFETTI’s within-tract spatial coupling to emerge, providing a showcase result for the proposed convolutional tract analysis with CONFETTI over traditional per-tract mean summaries.

Keywords: Graph Convolutional Network · Diffusion MRI · Fiber Tractography · Spatial Attribution · Down Syndrome.

1 Introduction

White matter (WM) microstructure analysis from diffusion MRI data typically reduces fiber bundles to one number per diffusion metric: a tract-average Fractional Anisotropy (FA), or Neurite Density Index (NDI) or similar [11]. This compression is convenient for traditional statistical analyses and classifiers [8,

7] but deliberately removes the along-tract spatial profile reducing the spatial localization of effects [10, 13]. Fiber-bundle methods such our atlas based fiber parametrization approach [13] or similar ones such the Automated Fiber Quantification family [10] generate along-tract, per-node diffusion metric profiles. Current profile analysis methods focus on per-tract analyses, and no all-tract/whole-brain methods exist that operates on these 1D profiles *jointly across tracts*, treating the whole white-matter representation as a single geometric object.

CONFETTI: We address this gap with **CONFETTI** (Convolutional Fiber Tract Inference), a multi-resolution graph convolutional pipeline for WM microstructure analysis combining (1) a hierarchical neighborhood graph over parametrized fiber axes with nested levels $L_0, \dots, L_5$ shared across subjects, that captures both within-tract continuity and between-tract proximity, (2) a per-subject implicit neural representation (INR) based imputation [9] for missing profiles, and (3) a GCN analysis stack with an out-of-fold interpretable attribution back-end (permutation importance [7], integrated gradients [6], and sparse GNNExplainer mask [5]). CONFETTI is architecture-agnostic; we compare here three structurally distinct GCN heads: a single-level GCN [1], a multi-scale parallel-branch concatenation [2, 3], and a hierarchical Graph U-Net [4].

Application: We evaluate CONFETTI in a study of WM microstructure in infants with Down Syndrome (DS), a cohort that offers a methodological stress test: per-tract mean baselines already saturate the classification AUC, so any added value from CONFETTI must come from *interpretation*. We benchmark CONFETTI against logistic regression and XGBoost on per-tract DTI/NODDI means under matched nested cross-validation on (a) DS classification and (b) prospective regression of 24-month cognitive and behavioral outcomes from 6-month imaging, then run focused validation experiments on the divergent finding.

Contributions: (i) **CONFETTI**, a multi-resolution graph convolutional pipeline for WM microstructure analysis with INR-based per-subject imputation and an out-of-fold GCN-based attribution stack. (ii) Three GCN instantiations (single-level, multi-scale concat, Graph U-Net) that share the same CONFETTI front-end and attribution back-end, making cross-architecture comparison meaningful. (iii) A DS showcase in which CONFETTI matches strong per-tract mean baselines in accuracy while the *anterior commissure* (AC) emerges as a top-ranked spatial finding that neither baseline lists in its highly ranked tracts; follow-up experiments establish the finding as biological signal rather than methodological artifact. (iv) A prospective-regression head showing that the same dominant spatial pattern predicts 24-month cognitive outcomes from 6-month imaging.

2 The CONFETTI Pipeline

2.1 From fiber tracts to multi-resolution graphs

Figure 1 summarizes the pipeline; the remainder of this section details each component in turn.

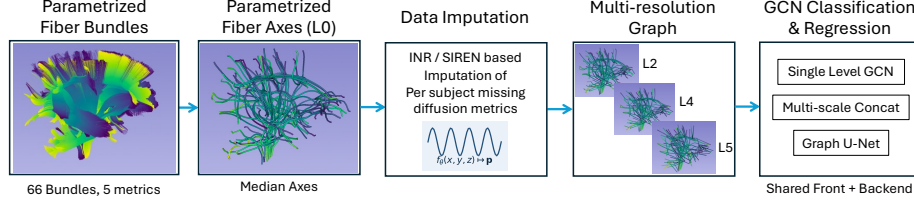


Fig. 1. CONFETTI overview. (a) Per-subject parametrized fiber bundles (66 tracts, 5 DTI/NODDI metrics); (b) per-tract parametrized fiber axes at L_0 , computed as a median location each bundle’s fibers; (c) SIREN per-subject INR imputation of missing profiles; (d) hierarchical multi-resolution neighborhood graph (L_2 , L_4 , L_5 shown); (e) three GCN heads sharing the CONFETTI front-end and attribution back-end.

For each of 66 pre-segmented white-matter tracts, we compute a 1D parametrized fiber axis by binning per-subject fiber points along their canonical arclength and taking the per-bin median, with end-clipping for outliers. A neighborhood graph is built over the concatenation of all axis points: two points are connected if they are adjacent on the same axis or within $1.5\times$ the median nearest cross-tract distance, with edge weights $\exp(-d^2/\sigma^2)$. From the finest graph L_0 ($\sim 4,766$ nodes) we recursively subsample to produce $L_1, \dots, L_5$ while preserving each tract’s endpoints. This hierarchy is computed once and shared across subjects.

2.2 Per-subject node features and INR imputation

Fiber QC procedures commonly remove 5-10% of fiber profiles resulting in uneven profile coverage which needs correction via data imputation. At each axis node we attach five diffusion metrics (FA, RD, AD, NDI, ODI) and four spatial features (x, y, z, ℓ) for a 9-D input. Any missing profile data were imputed using an implicit neural representation (INR) model using SIREN [9] activation functions (200 epochs, $\omega_0 = 10$). Per-fold training-set statistics are used so that the imputation respects the nested-CV split. Three subject-level covariates (sex, gestational age at birth, DWI artifact level) are mean-imputed per fold and concatenated to the graph embedding before the MLP head, keeping demographics separable from the spatial pattern under attribution. We compare the INR imputation versus traditional baselines (kNN with 5 and 10 neighbors, linear, mean, median imputation).

2.3 GCN architectures for CONFETTI

CONFETTI is architecture-agnostic at the head; we instantiate it with three GCN architectures whose inductive biases differ significantly. **Single-level GCN** is a two-layer Kipf-Welling GCN at a single, relatively coarse level only (here L_3). **Multi-scale concat** runs an independent two-layer GCN at two levels (here L_2 and L_3) in parallel and concatenates global-pooled embeddings before the

Table 1. Down Syndrome classification: held-out CV AUC and F1 with covariates concatenated at the head. Baselines compute per-tract DTI/NODDI means; CONFETTI uses per-node features under the multi-resolution graph pipeline. Numbers are mean $\pm$ std over 15 outer folds (5×3 repeats), best dropout per model.

Method	AUC	F1
LogReg (per-tract means)	1.000 ± 0.000	0.994 ± 0.015
XGBoost (per-tract means)	0.983 ± 0.018	0.921 ± 0.052
CONFETTI - Single-level GCN (L3)	0.913 ± 0.125	0.809 ± 0.250
CONFETTI - Multi-scale concat (L2+L3)	0.931 ± 0.084	0.820 ± 0.128
CONFETTI - Graph U-Net (L2-L4)	0.940 ± 0.106	0.888 ± 0.093

head. **Hierarchical Graph U-Net** replaces dynamic pooling with the fixed index hierarchy above (from L2 to L3 to L4): a GCN block per encoder level, scatter-unpool with skip connections in the decoder, and global mean+max pool of the finest decoder level into the output head. Best models are selected on inner-fold validation AUC (or negative-MAE for regression).

2.4 Attribution stack

For each architecture we compute three complementary attributions: out-of-fold permutation importance [7] for tracts, per-node positions, covariates, and (in regression) per-target; integrated gradients on the input feature tensor [6]; and a sparse GNNExplainer mask [5] per held-out subject. The permutation importance metric is the held-out AUC (or difference in held-out MAE for regression) when a feature group is shuffled across subjects in the outer-CV test fold. Crucially this is computed inside the nested-CV loop, so the model used for attribution has never seen the test subjects.

3 Application: Infant Down Syndrome Study

3.1 Dataset and cross-validation protocol

We evaluate on an infant Down Syndrome (DS) cohort [13] with imaging at 6 months and outcome assessment at 24 months with 59 infants with DS and 68 control infants. Five diffusion metrics are assessed: FA, RD (Radial Diffusivity), AD (Axial Diffusivity), NDI, and ODI (Orientation Dispersion Index). For the prospective regression we predict scores from two behavioral/cognitive assessments measured at 24 months of age: Vineland-III Adaptive Behavior Composite, Communication, Daily Living Skills, Motor, and Socialization scores; as well as Bayley-4 Cognitive standard, Expressive and Receptive Communication scores.

All experiments use a stratified 5-fold cross-validation repeated 3 times with inner-fold model selection.

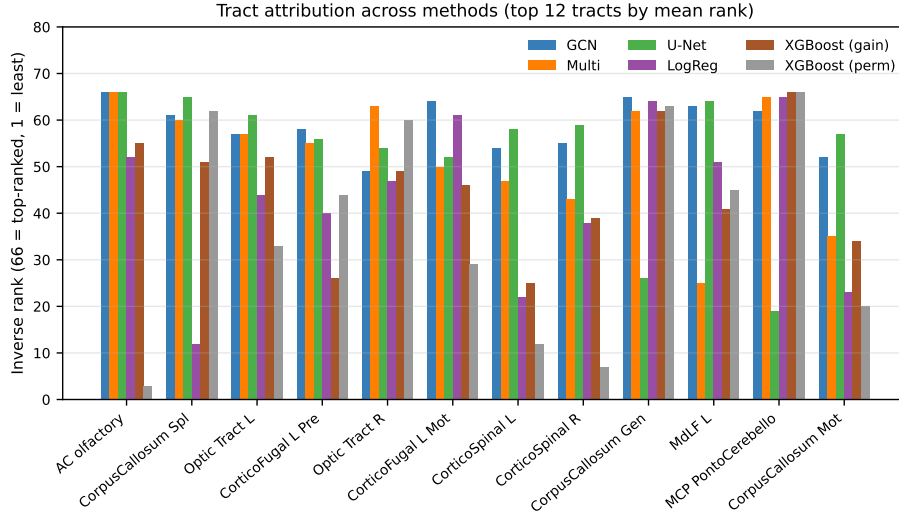


Fig. 2. Cross-method tract attribution. Per-method inverse rank (66 minus rank) for the top tracts. The three CONFETTI models agree on the AC olfactory at rank 1; the baselines rank the MCP and CC Lateral Genu tracts highest. The MCP and CC Lateral Genu tracts appear in $\geq 5/6$ top-10 lists; AC olfactory is CONFETTI-specific.

3.2 Imputation evaluation

We evaluated the INR imputation on a held-out subset of 34 subjects with complete data by removing 7 random tracts per subject and reconstructing them from the rest. Normalized mean absolute error (NMAE) was lowest for INR/SIREN (0.520), followed by kNN-10 (0.576, +11% vs INR), kNN-5 (+16%), linear (+30%), median (+34%), and mean (+37%). The INR/SIREN imputation outperforms all baselines significantly ($p < 0.001$).

3.3 Classification: CONFETTI is close to baselines, but with different attribution

Table 1 summarizes the held-out classification performance. Logistic regression on per-tract means saturates at a near perfect performance $AUC = 1.000$, $F1 = 0.994$; XGBoost reaches 0.983/0.921. All three CONFETTI models land a bit lower, between 0.913 and 0.953, with the Graph U-Net at the upper end. The gap (at this sample size) is relatively small, likely due to the larger complexity of the GCN models.

Per-tract attribution tells a different story for the baselines and the GCN models. Figure 2 compares the inverse rank (highest bar = top-ranked tract) of the top ranking tracts. The two baselines put the *Mid Cerebellar Pontine (MCP) tract* at rank 1, with the *Corpus Callosum (CC) Lateral Genu tract* a close second. In contrast, all three CONFETTI permutation rankings put the

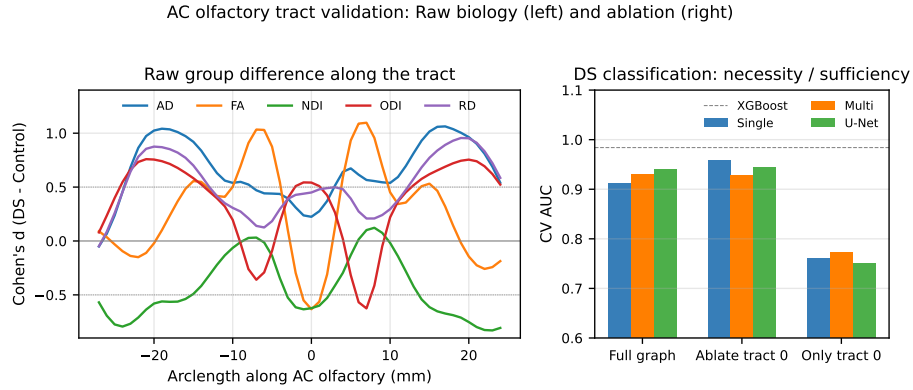


Fig. 3. Validation of the AC olfactory finding. Left: per-node Cohen’s d along AC olfactory for each diffusion property; horizontal dotted lines mark $|d| = 0.5$. Right: classification AUC with the full graph, AC olfactory ablated, and only AC olfactory retained; dashed line is XGBoost.

Anterior Commissure (AC) olfactory at rank 1 by a wide margin, while still flagging the MCP tract as a secondary contributor. Furthermore, spearman rank correlation across all 66 tracts is 0.64 between the two baselines, but only (at most) 0.34 between any CONFETTI instantiation and any baseline.

3.4 Validating CONFETTI’s divergent finding

We ran five experiments to establish whether CONFETTI’s AC olfactory tract importance is a biological signal or a methodological artifact.

(1,2) *Raw biology and per-property channels.* Welch two-sample t -tests at each of the 52 AC olfactory nodes per property (260 tests, FDR-corrected with Benjamini-Hochberg [16]) yield 174/260 significant at $q < 0.05$, and all 52 nodes are significant for at least one property; peak per-node Cohen’s d reaches +1.10 for AD and +1.09 for FA (Fig. 3, left). While the overall direction (AD $\uparrow$, RD $\uparrow$, NDI $\downarrow$, ODI $\uparrow$) matches the hypomyelination + reduced-neurite-density signature reported in DS cohorts [13, 15, 14], the group differences invert direction for FA, ODI, and NDI. This effect inversion cannot be captured by tract means and is likely one of the reasons the baselines do not rank the AC olfactory highly.

Restricting the OOF permutation to AC olfactory and shuffling one property at a time isolates AD (+0.048 gcn, +0.036 unet) and RD as the predictive drivers; NDI and ODI contribute near zero. CONFETTI’s attribution at the AC olfactory tract therefore loads on the axial and radial diffusivity channels.

(3,4) *Necessity and sufficiency.* Replacing AC olfactory’s per-node values with the training-fold mean does not hurt classification (CV AUC 0.913 $\rightarrow$ 0.959 gcn,

Table 2. 6 month imaging predicting 24 month outcomes. Pearson r on held-out folds (mean over 3x5 folds; std in parentheses). **Bold** = best of row.

V24 outcome	Ridge baseline	XGBoost baseline	Graph U-Net
Vineland ABC	0.37 (.11)	0.46 (.12)	0.53 (.11)
Vineland Communication	0.44 (.14)	0.53 (.13)	0.54 (.16)
Vineland Daily Living Skills	0.32 (.19)	0.48 (.12)	0.51 (.13)
Vineland Motor	0.64 (.13)	0.68 (.11)	0.66 (.12)
Vineland Socialization	0.18 (.12)	0.22 (.13)	0.39 (.20)
Bayley Cognitive	0.55 (.13)	0.65 (.11)	0.63 (.15)
Bayley Expressive Comm.	0.44 (.11)	0.51 (.11)	0.53 (.16)
Bayley Receptive Comm.	0.64 (.11)	0.62 (.10)	0.58 (.13)

0.931 $\rightarrow$ 0.929 multi, 0.940 $\rightarrow$ 0.944 unet): the models redistribute information across the remaining 65 tracts, consistent with the expected brain-wide DS pathology. Training on AC olfactory alone reaches 0.761/0.774/0.752 (gcn/multi/unet), well above chance from the ≤ 52 axis points (Fig. 3, right). AC olfactory is therefore informative but redundant for classification.

(5) *Within-tract spatial peak.* Aggregating the OOF GNNExplainer per-DS-subject masks places the spatial peak at arclength +13mm across all three CONFETTI architectures, with $\geq 1.6\times$ activation versus other-tract nodes. This convergence under a different attribution mechanism than (1-4) indicates a likely stable spatial signal rather than a permutation artifact.

3.5 Predictive regression: 6-month imaging predicts later outcomes

We applied CONFETTI to a multi-output regression on all 24-month outcomes, training jointly (z-score targets, smooth L1 loss, per-target NaN masking). Table 2 shows held-out Pearson r for the Graph U-Net instantiation (the strongest CONFETTI head on this task) against the two baselines. CONFETTI outperforms both baselines on 5 of 8 outcomes and is within 0.05 of XGBoost on the rest. The largest absolute Pearson r is on Bayley COG and Vineland Motor.

Per-target attribution. Figure 4 shows CONFETTI’s per-24-month-target OOF tract permutation importance for the Graph U-Net head. AC olfactory is the top-attributed tract for all outcomes. The single largest cell in the heatmap is AC olfactory $\times$ Bayley Cognitive where permuting AC olfactory tract features increases the held-out MAE on the Bayley Cognitive score by approximately 0.80. Aggregating across the three CONFETTI architectures, AC olfactory ranks first in 22 of 24 (target $\times$ architecture) combinations.

It worthwhile to note that the left and right optic tract are consistent secondary contributors. These tracts show an opposite directional signature to the AC olfactory tract: higher FA and lower RD/ODI indicating a "more coherent / better organized microstructure" pattern. This finding is biologically surprising, possibly reflecting altered visual-system development in DS.

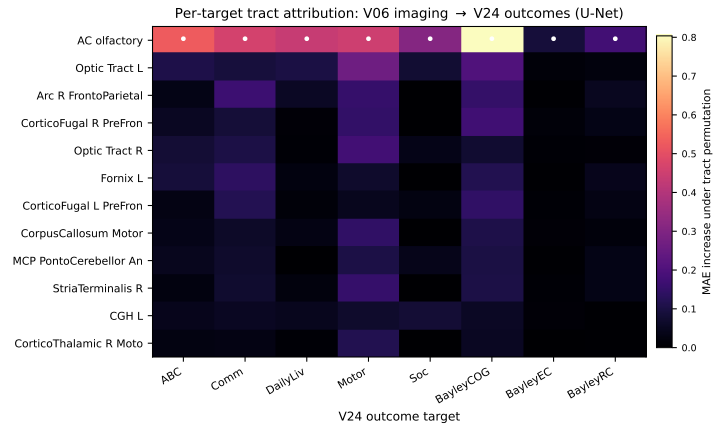


Fig. 4. CONFETTI per-target tract attribution for the 6-month-diffusion→24-month-behavior/cognition predictive regression (Graph U-Net). Cell color is the held-out MAE increase under row-tract permutation; bullets mark the top tract per column. AC olfactory tops every column.

The same OOF protocol applied to the three covariates shows that gestational age shows a predictive signal for the Bayley Cognitive score (+0.264 MAE), and sex for the Daily Living score (+0.212). The AC olfactory’s attribution significantly exceeds covariate contributions.

4 Discussion

Under matched out-of-fold cross-validation, CONFETTI’s top-ranked tract differs from that of the per-tract-mean baselines: CONFETTI puts the AC olfactory tract at rank 1, while the tract mean methods put the middle cerebellar peduncle at rank 1 with AC olfactory only mid-list. The follow-up experiments establish this as biological signal rather than artifact, and the same tract dominates the predictive outcome regression. The likely reasons the baselines miss this finding: (1) group differences invert direction along the tract which cannot be captured by tract means, (2) the AC olfactory has also particularly sharp group differences that mean-pooling averages away. The architecture-independent consensus tracts (CC GenuLateral, MCP PontoCerebellar Anterior, CC Splenium) remain visible in CONFETTI’s attribution.

5 Conclusion

Our novel fiber neighborhood graph based analysis framework called **CONFETTI** recovers spatial signals that traditional per-tract-mean baselines do not reveal with robust, novel findings of the relevance of the AC olfactory tract in DS.

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